Bora Ayhan

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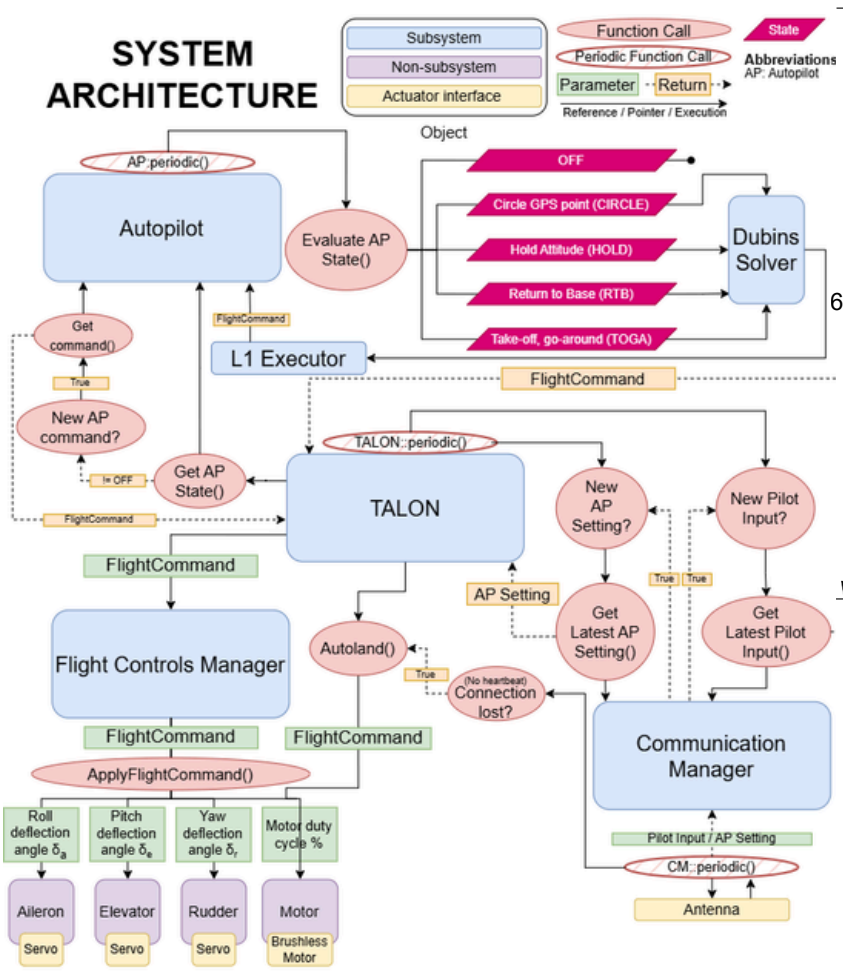
Trim, Auto Land, Odometry, and Navigation (TALON) Flight Computer **Jun. 2026 – Aug. 2026**

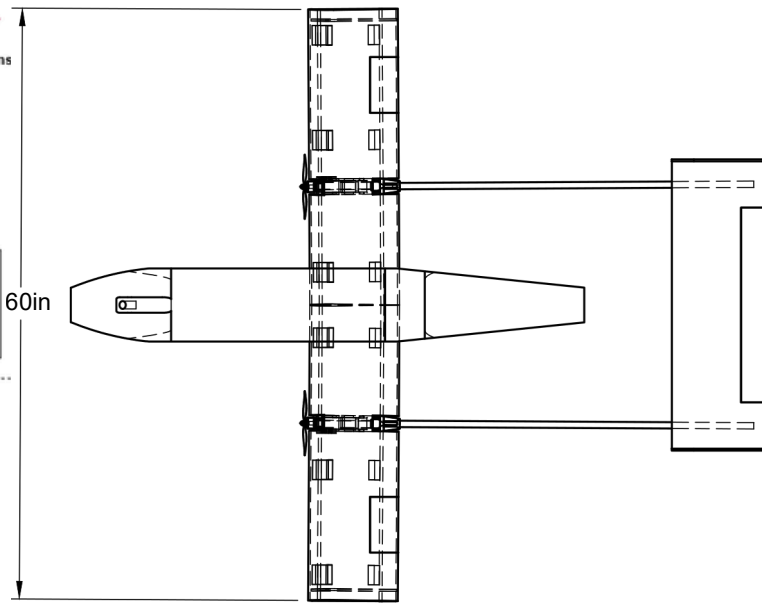
Creation of a C++ flight controller with autopilot functionality for Class 1 UAVs

- Architected minimally coupled flight software, isolating guidance and input logic from actuator subsystem.
- Implements Dubins path generation and L1 guidance (Park, Deyst, & How, 2004) for flight path execution.

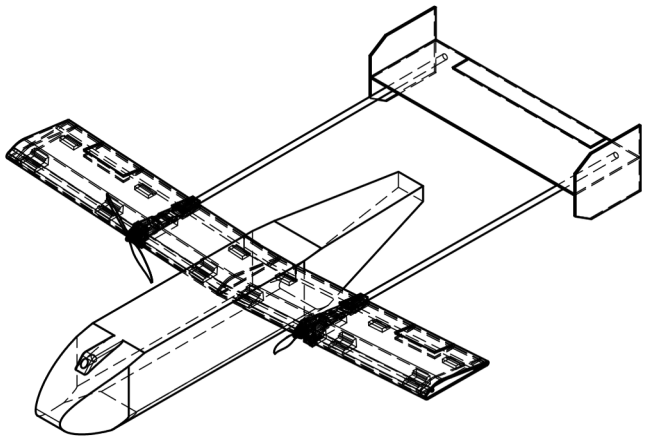
Subproject: Design and manufacture of a dual propeller fixed wing UAV testbed for evaluating TALON

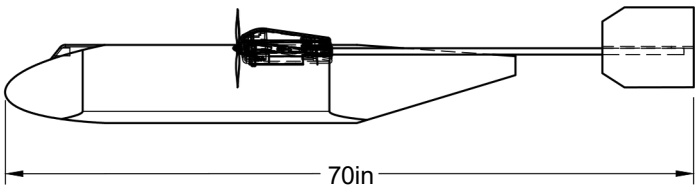
- 60-in wingspan, <$45/unit, positively longitudinally stable drone with 3kg MTOW
- 3D-printed wingbox and rib structure to maintain Clark-Y airfoil at 5° dihedral for roll stability.
- Integrated wing-embedded assembly (80 g weight) joining motor, ESC, battery, and tail boom.
- Electronics and radio embedded entirely within airfoil structure to reduce parasitic drag.











Videos of TALON in flight are available upon request. Please reach out to the phone number above.